MACHINE LEARNING

COMPLETE VIVA QUESTIONS & ANSWERS

PCCST503 | S5 CSE | Modules 1-4

Student revision handbook | Concise oral answers, key definitions and formulas

Module 1 - Introduction, Parameter Estimation And Regression

49 viva questions

Module 2 - Classification And Evaluation

62 viva questions

Module 3 - Svm And Neural Networks

40 viva questions

Module 4 - Clustering, Dimensionality Reduction And Ensembles

56 viva questions

Study tip: Answer each question aloud before reading its answer. Revise algorithm steps, formulas, and worked problems separately.

MODULE 1 - INTRODUCTION, PARAMETER ESTIMATION AND REGRESSION

1. What is Machine Learning?

Machine Learning is a branch of AI in which models learn patterns from data to make predictions or decisions without hand-coding every rule.

2. How does ML differ from traditional programming?

Traditional programming applies human-written rules to inputs; ML learns model parameters or patterns from training examples.

3. Name four learning paradigms.

Supervised, unsupervised, semi-supervised, and reinforcement learning.

4. What is supervised learning?

Learning from examples containing both input features and known targets; classification and regression are examples.

5. What is unsupervised learning?

Finding structure in data without supplied target labels, such as clusters or lower-dimensional representations.

6. What is semi-supervised learning?

Learning from a mixture of a small labelled dataset and a larger unlabelled dataset.

7. What is reinforcement learning?

An agent learns actions through interaction with an environment to maximize expected cumulative reward.

8. What is classification?

Prediction of a discrete class, such as pass/fail or spam/not spam.

9. What is regression?

Prediction of a numerical target, such as price or temperature.

10. What is a dataset?

A collection of observations used for learning, evaluating, or analyzing a model.

11. What is a feature?

An input variable or attribute describing an observation.

12. What is a label or target?

The known outcome the model aims to predict in supervised learning.

13. What is a trained model?

A learned mapping from input features to predicted outputs.

14. What is training?

The process of fitting a model's parameters to data using a learning objective.

15. What is parameter estimation?

Determining unknown model parameters from observed data.

16. What is a parameter?

A quantity learned from data, such as a regression coefficient or intercept.

17. What is a hyperparameter?

A configuration selected outside ordinary parameter fitting, such as learning rate or K in KNN.

18. What is Maximum Likelihood Estimation (MLE)?

Choosing parameter theta that maximizes P(D | theta), the likelihood of the observed dataset D.

19. What is likelihood?

A function of the model parameter that expresses how compatible the observed data is with that parameter.

20. Why use log-likelihood?

Logarithms turn products into sums, simplify differentiation, and often improve numerical stability.

21. What is Maximum A Posteriori (MAP) estimation?

Choosing theta that maximizes posterior P(theta | D), incorporating both data and prior beliefs.

22. State Bayes' theorem for parameter estimation.

P(theta | D) = P(D | theta) P(theta) / P(D).

23. What is a prior?

A probability distribution describing beliefs about a parameter before observing the current data.

24. What is a posterior?

The updated distribution over a parameter after incorporating observed data.

25. What is evidence in Bayes' theorem?

P(D), the marginal probability of the observed data; it normalizes the posterior.

26. What is the difference between MLE and MAP?

MLE maximizes likelihood alone; MAP maximizes likelihood multiplied by the prior.

27. When can MLE and MAP coincide?

When the prior is constant across the relevant parameter domain, their maximizers can coincide.

28. What is Bayesian learning?

A probabilistic approach that updates distributions over unknown quantities using observed data.

29. What is feature representation?

Converting raw observations into useful numerical or categorical input variables for a model.

30. What is ML problem formulation?

Defining the inputs, target, learning task, objective, assumptions, and evaluation measure.

31. What is a loss function?

A function quantifying disagreement between predictions and target values.

32. What is an objective function?

The quantity optimized during training, often prediction loss plus regularization.

33. What is optimization?

Searching for parameters that minimize or maximize an objective function.

34. What is gradient descent?

An iterative optimization method that updates theta <- theta - eta * gradient J(theta).

35. What is learning rate?

The step-size eta used in parameter updates.

36. What if learning rate is too high?

Optimization can overshoot minima, oscillate, or diverge.

37. What if learning rate is too low?

Optimization may be very slow.

38. What is simple linear regression?

A numerical prediction model with one input: y_hat = w0 + w1*x.

39. What is multiple linear regression?

A numerical prediction model with multiple inputs: y_hat = w0 + w1*x1 + ... + wn*xn.

40. What is the slope in simple linear regression?

The predicted change in output for a one-unit increase in input.

41. What is the intercept?

The predicted output when all input features equal zero.

42. What is Mean Squared Error (MSE)?

The mean of squared prediction errors: (1/n) * sum((y_i - y_hat_i)^2).

43. What is the ordinary least squares objective?

Choose coefficients minimizing the sum of squared residuals.

44. What is the normal equation?

If X'X is invertible, w_hat = (X'X)^(-1) X'y; a pseudoinverse is useful otherwise.

45. Gradient descent versus matrix method?

Gradient descent updates coefficients iteratively; matrix-based least squares solves a linear algebra problem.

46. Why scale features before gradient descent?

Comparable scales can improve the conditioning and speed of optimization.

47. Batch versus stochastic gradient descent?

Batch uses all training examples per update; stochastic uses one example per update.

48. What is a residual?

The difference between an observed target and its predicted value, y - y_hat.

49. What is training error versus test error?

Training error is measured on fitted examples; test error evaluates performance on held-out data.

MODULE 2 - CLASSIFICATION AND EVALUATION

50. What is binary classification?

Predicting one of two possible class labels.

51. What is multiclass classification?

Predicting one of three or more mutually exclusive classes.

52. What is Logistic Regression?

A classifier that models a class probability using a logistic link applied to a linear score.

53. What is the sigmoid function?

sigmoid(z) = 1/(1 + exp(-z)); its output lies strictly between zero and one for finite z.

54. Why use sigmoid in binary classification?

It maps a real-valued score to a number interpretable as a modelled positive-class probability.

55. What is a classification threshold?

A cutoff used to turn a probability or score into a class prediction; 0.5 is common but not universal.

56. Why is Logistic Regression a classifier?

Its usual target is a class probability and corresponding categorical decision, despite its name.

57. What is Naive Bayes?

A probabilistic classifier applying Bayes' theorem with conditional independence of features given the class.

58. Why is Naive Bayes called naive?

It makes the simplifying assumption that features are conditionally independent given the class.

59. State Bayes' theorem for classification.

P(C | X) = P(X | C) P(C) / P(X).

60. What is a class prior?

P(C), the probability of a class before considering the current observation's features.

61. What is class likelihood?

P(X | C), the likelihood of observed features given a class.

62. What is K-Nearest Neighbours (KNN)?

A supervised method that predicts using the targets of the K closest training examples.

63. Is standard KNN supervised?

Yes; KNN classification and regression require labelled training examples.

64. What does K mean in KNN?

The number of nearest training observations used for a prediction.

65. What happens when K = 1?

The closest example alone determines the prediction; results can be sensitive to noise.

66. Why scale inputs for KNN?

Distances can be dominated by features measured on larger numerical scales.

67. What is Euclidean distance?

The square root of the sum of squared coordinate differences.

68. What is a decision tree?

A predictive model that recursively tests features and ends at leaves with predictions.

69. What is a root node?

The first decision node at the top of the tree.

70. What is a leaf node?

A terminal node that outputs a class or predicted value.

71. What is ID3?

A decision-tree algorithm that chooses splitting attributes using information gain based on entropy.

72. What is entropy in classification?

A measure of class impurity: H(S) = -sum(p_i log2(p_i)).

73. What is information gain?

Parent entropy minus the weighted average entropy of the child groups.

74. How does ID3 choose an attribute?

It selects the available attribute with the highest information gain at a node.

75. When does a decision tree stop splitting?

For example when a node is pure, attributes are exhausted, or a stopping criterion is met.

76. What is generalization?

Ability to perform well on relevant unseen examples.

77. What is overfitting?

Learning training-specific noise or details that do not generalize well.

78. What is underfitting?

Failing to capture important structure, often because the model is too simple.

79. What is a good fit?

A model that captures useful structure while maintaining good performance on unseen data.

80. Typical errors in overfitting?

Training error is often low while validation/test error is comparatively high.

81. Typical errors in underfitting?

Both training and test errors are often high.

82. What is regularization?

Penalizing complexity or constraining a model to improve generalization.

83. What is LASSO?

L1 regularization: loss + lambda * sum(abs(w_j)); it can set some coefficients to zero.

84. What is Ridge regression?

L2 regularization: loss + lambda * sum(w_j^2); it shrinks coefficients toward zero.

85. What is lambda in regularization?

A hyperparameter controlling the strength of the penalty.

86. LASSO versus Ridge?

LASSO uses absolute coefficients and may select features; Ridge uses squared coefficients and usually retains them.

87. What is training data?

Examples used to fit model parameters.

88. What is validation data?

Held-out examples used during development for model choice and hyperparameter tuning.

89. What is test data?

Held-out examples used for final evaluation after model selection.

90. Why not tune repeatedly on the test set?

It leaks information into model selection and can give overly optimistic performance estimates.

91. What is a confusion matrix?

A table comparing predicted labels with actual labels, including TP, TN, FP, and FN in binary classification.

92. What is a true positive?

An actually positive observation predicted positive.

93. What is a true negative?

An actually negative observation predicted negative.

94. What is a false positive?

An actually negative observation incorrectly predicted positive.

95. What is a false negative?

An actually positive observation incorrectly predicted negative.

96. What is accuracy?

Accuracy = (TP + TN)/(TP + TN + FP + FN).

97. What is precision?

Precision = TP/(TP + FP), when the denominator is nonzero.

98. What is recall?

Recall = TP/(TP + FN), when the denominator is nonzero.

99. What is F1-score?

F1 = 2 * precision * recall/(precision + recall), when defined.

100. When prioritize precision?

When false positives are especially costly.

101. When prioritize recall?

When missing positive cases is especially costly.

102. What is an ROC curve?

A plot of true-positive rate against false-positive rate across decision thresholds.

103. What is true-positive rate?

TPR = TP/(TP + FN); also called recall or sensitivity.

104. What is false-positive rate?

FPR = FP/(FP + TN).

105. What is AUC?

Area under the ROC curve; it summarizes ranking discrimination across thresholds.

106. ROC versus AUC?

ROC is the curve; AUC is a scalar summary of its area.

107. What is MAE?

Mean Absolute Error = (1/n) * sum(abs(y_i - y_hat_i)).

108. What is RMSE?

Root Mean Squared Error = sqrt((1/n) * sum((y_i - y_hat_i)^2)).

109. What is R-squared?

R2 = 1 - sum((y_i-y_hat_i)^2)/sum((y_i-y_mean)^2), when the denominator is nonzero.

110. MAE versus RMSE?

RMSE penalizes large errors more strongly because errors are squared before averaging.

111. What does negative R-squared mean?

The model has worse squared-error performance than predicting the evaluation target mean.

MODULE 3 - SVM AND NEURAL NETWORKS

112. What is a Support Vector Machine?

A supervised model that learns a maximum-margin decision boundary for classification.

113. What is a hyperplane?

A linear decision boundary described by w' x + b = 0.

114. What is a maximum-margin hyperplane?

The separating boundary maximizing the margin under the chosen SVM formulation.

115. What is an SVM margin?

The distance from the separating boundary to the nearest class points in the canonical hard-margin setting.

116. What are support vectors?

Training examples that determine or influence the learned separating boundary.

117. What is the full canonical SVM margin width?

The distance between w' x + b = +1 and -1 is 2/||w||.

118. Why maximize margin?

A wider margin can improve robustness and generalization.

119. What is hard-margin SVM?

An SVM requiring linearly separable data with no margin violations.

120. What is soft-margin SVM?

An SVM allowing violations with slack variables and a penalty on them.

121. What does C control in soft-margin SVM?

The penalty strength for margin violations; higher C penalizes violations more.

122. What is a nonlinear SVM?

An SVM using a suitable kernel to learn nonlinear boundaries in the original feature space.

123. What is the kernel trick?

Computing feature-space inner products through a kernel without explicitly constructing the mapping.

124. Name common SVM kernels.

Linear, polynomial, and radial basis function (RBF) kernels.

125. What is the RBF kernel?

K(x,z) = exp(-gamma * ||x-z||^2), with gamma greater than zero.

126. Linear versus nonlinear SVM?

Linear SVM uses a linear original-space boundary; kernel SVM can yield nonlinear boundaries.

127. SVM versus KNN?

SVM learns a separating boundary; KNN predicts using nearby stored training examples.

128. What is an ANN?

A computational model of interconnected artificial neurons that learns input-output mappings.

129. What is a perceptron?

A neuron or simple linear-threshold classifier applying activation to a weighted input sum.

130. What is a neuron's basic equation?

z = sum(w_i*x_i) + b; output a = f(z).

131. What is a weight?

A learned coefficient controlling an input's contribution to a neuron.

132. What is a bias?

An adjustable offset added to the weighted sum.

133. What is a single-layer perceptron?

An input connected directly to a trainable output layer with no hidden layer.

134. What is a multilayer perceptron?

A feed-forward network with one or more hidden layers and an output layer.

135. Why use hidden layers?

They learn intermediate representations; nonlinear activations enable complex nonlinear mappings.

136. SLP versus MLP?

SLP has no hidden layer; MLP has at least one hidden layer.

137. What is an activation function?

A function transforming a neuron's weighted sum into its output.

138. Why are nonlinear activations necessary?

Without them, stacked affine layers collapse to one affine transformation.

139. What is sigmoid?

sigmoid(z) = 1/(1 + exp(-z)), mapping real numbers into (0,1).

140. What is ReLU?

ReLU(z) = max(0,z).

141. What is tanh?

An activation mapping real inputs to values strictly between -1 and +1.

142. Sigmoid versus tanh?

Sigmoid maps to (0,1); tanh maps to (-1,1).

143. What is forward propagation?

Computing the network's output from input through successive layers.

144. What is neural-network loss?

A measure of the difference between the predicted output and target.

145. What is backpropagation?

Computing loss gradients through layers in reverse order using the chain rule.

146. Backpropagation versus gradient descent?

Backpropagation calculates gradients; gradient descent uses gradients to update weights.

147. What is an epoch?

One full pass through the training dataset.

148. What is a batch?

A subset of training examples used for one optimization step.

149. What is the vanishing gradient problem?

Gradients become very small through layers, slowing learning in earlier layers.

150. Can MLP solve XOR?

Yes, with a suitable hidden layer and nonlinear activations; a standard single-layer perceptron cannot.

151. How many output neurons are needed?

It depends on the task: commonly one sigmoid for binary classification or K softmax outputs for K mutually exclusive classes.

MODULE 4 - CLUSTERING, DIMENSIONALITY REDUCTION AND ENSEMBLES

152. What is clustering?

Grouping observations by similarity without requiring class labels.

153. What is a similarity measure?

A numerical way to quantify how alike observations are.

154. What is Euclidean distance?

d(x,y) = sqrt(sum((x_i-y_i)^2)).

155. What is Manhattan distance?

d(x,y) = sum(abs(x_i-y_i)).

156. What is hierarchical clustering?

Building a nested hierarchy of clusters, often visualized as a dendrogram.

157. What is agglomerative clustering?

Bottom-up hierarchical clustering that repeatedly merges clusters.

158. What is divisive clustering?

Top-down hierarchical clustering that repeatedly splits groups.

159. What is a dendrogram?

A tree diagram showing cluster mergers or splits and their associated levels.

160. What is linkage?

The rule used to define distance between two clusters.

161. What is single linkage?

The minimum distance between a point in each of two clusters.

162. What is complete linkage?

The maximum pairwise distance between points in two clusters.

163. What is average linkage?

The average pairwise distance between points in two clusters.

164. What is K-Means?

An algorithm that partitions observations into K groups by repeatedly assigning points to nearest centroids and updating centroids.

165. What is a centroid?

The arithmetic mean feature vector of observations in a cluster.

166. What is the K-Means objective?

Minimize sum over clusters and points of squared Euclidean distances to assigned centroids.

167. What are K-Means steps?

Choose K; initialize centroids; assign points; recompute centroids; repeat until stopping.

168. How can K be chosen?

Elbow analysis, silhouette scores, and domain knowledge can guide the choice.

169. What is inertia?

The within-cluster sum of squared distances to assigned centroids.

170. Why does inertia usually fall as K rises?

More centroids allow a lower or equal optimal within-cluster sum of squares.

171. What is silhouette score?

A cohesion/separation score in [-1,1]; higher values generally indicate better-separated clusters.

172. Hierarchical clustering versus K-Means?

Hierarchical builds nested clusters; K-Means directly partitions into a specified K groups.

173. What is dimensionality reduction?

Representing observations with fewer features while retaining useful information.

174. Why reduce dimensionality?

To simplify visualization, reduce computation, or limit redundancy/noise.

175. What is PCA?

Principal Component Analysis, a linear projection onto orthogonal directions of maximum variance.

176. What is a principal component?

A direction formed as a linear combination of original features.

177. What is the first principal component?

The direction capturing the greatest variance of centered data.

178. What is explained variance?

The variance accounted for by a component or selected set of components.

179. What are PCA steps?

Center data, obtain covariance/equivalent decomposition, rank principal directions, and project onto chosen directions.

180. Does PCA require labels?

No, standard PCA is unsupervised.

181. Does PCA always improve accuracy?

No; it may discard predictive information.

182. What is multidimensional scaling (MDS)?

A method embedding objects in fewer dimensions while approximating pairwise distances or dissimilarities.

183. PCA versus MDS?

PCA preserves high-variance directions; MDS aims to preserve pairwise dissimilarities.

184. What is an ensemble method?

Combining multiple predictive models to improve accuracy, stability, or generalization.

185. Why use ensembles?

They can reduce variance and combine complementary model strengths.

186. What is bagging?

Bootstrap aggregating: train learners on resampled datasets and aggregate their predictions.

187. What is bootstrapping?

Sampling observations with replacement from an existing dataset.

188. What is boosting?

Sequentially combining learners, where later stages address earlier errors or residuals.

189. Bagging versus boosting?

Bagging fits models independently on bootstrap samples; boosting adds models sequentially.

190. Give an example of bagging.

Random Forest is a bagging-based ensemble of randomized decision trees.

191. Give an example of boosting.

AdaBoost and Gradient Boosting.

192. Does bagging reduce bias or variance?

It primarily reduces variance for unstable base learners.

193. Can boosting reduce bias?

Yes; depending on the method, boosting can reduce bias and sometimes variance.

194. Single tree versus Random Forest?

A single tree predicts alone; Random Forest aggregates predictions of many randomized trees.

195. What is resampling?

Repeatedly sampling or splitting observed data to assess variability or model performance.

196. What is cross-validation?

Repeatedly training and validating across different partitions of the available data.

197. What is K-fold cross-validation?

Divide data into K folds; each fold is used once for validation while others train the model.

198. Why use cross-validation?

To estimate generalization and support model/hyperparameter selection.

199. Cross-validation versus bootstrapping?

CV rotates held-out folds; bootstrap draws replacement samples.

200. What is bias?

Systematic error associated with simplifying assumptions or insufficient model flexibility.

201. What is variance?

Sensitivity of a learned model to fluctuations in the training sample.

202. What is bias-variance trade-off?

Balancing underfitting and sensitivity to training data to achieve good generalization.

203. Underfitting is associated with what?

Typically high bias.

204. Overfitting is associated with what?

Typically high variance.

205. How can overfitting be reduced?

Regularization, simpler models, suitable validation, early stopping, and representative data.

206. How can underfitting be reduced?

Use more suitable features or models, reduce excessive regularization, or train sufficiently.

207. Why fit preprocessing on training data only?

To avoid data leakage from validation/test data into training.

FINAL VIVA CHECKLIST

• Explain the task each algorithm solves and whether it is supervised or unsupervised.

• Explain the major steps of each algorithm in the correct order.

• Recall the important formulas and define their symbols.

• Give a simple application, one advantage, and one limitation.

• Practise numerical problems: gradient descent, Bayes, entropy/information gain, metrics, SVM margin, K-Means, and PCA.